

Cloudy with a Chance of Compute

Mapping the hidden geography of AI compute across 8,000 cities — and who it leaves behind

Matthew Tan

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Introduction

AI does not happen on a blank map.

Modern AI systems depend on physical cloud compute infrastructure to train models, serve real-time inference, and support the applications that drive AI adoption. This infrastructure — the data center regions operated by AWS, Azure, and Google Cloud — is concentrated in specific locations, primarily in North America, Europe, and parts of East Asia. Distance to these regions affects network latency, data transfer costs, and the practical viability of running AI workloads at scale: industry research estimates that network-induced latency alone adds 50 to 150 milliseconds per request for cities far from centralized cloud infrastructure, frequently exceeding the thresholds required for responsive AI applications. For a city like Singapore, which sits 4 km from its nearest major cloud compute region, these systems are immediate and accessible. For a city like Lagos — one of the largest and fastest-growing urban centers on the African continent — the nearest major cloud region is nearly 3,800 km away. Both cities have significant scale, growing digital economies, and stated national ambitions in artificial intelligence. They do not begin from the same infrastructure position.

The scale of this imbalance is significant, and it is widening. In 2026, global hyperscaler capital expenditure on data center construction is projected to exceed \$600 billion, yet the distribution of that investment remains deeply uneven. Across the entire African continent — home to 1.4 billion people and 18 percent of the global population — less than 1 percent of global data center capacity currently resides. Only South Africa hosts major cloud regions from all three leading providers. No AWS, Azure, or Google Cloud region currently operates in Nigeria, in East Africa's largest economies, or anywhere in North or West Africa. A population larger than that of Europe or the Americas is served by a single country's compute footprint. The International Monetary Fund has identified access to computing power and data infrastructure as a critical determinant of which countries benefit from AI, and recent assessments indicate that AI adoption in the Global North is growing nearly twice as fast as in the Global South — a gap that multiple international bodies have warned risks deepening cross-country inequality rather than narrowing it.

Despite the growing recognition of this divide, public discussion of AI advantage continues to emphasize talent pipelines, venture capital, and national policy frameworks. The physical infrastructure layer — where compute capacity is actually located, how far cities sit from it, and whether that distance corresponds with observable differences in AI activity — remains largely absent from the conversation. There is, to date, no systematic global mapping of how cloud compute proximity relates to the geographic distribution of AI research.

This atlas addresses that gap. It measures the distance from each of the world's 8,000 largest cities to its nearest major cloud compute region, overlays observed AI research activity drawn from the OpenAlex scholarly record, and investigates a central question: once city size and geography are accounted for, do cities that are farther from cloud compute infrastructure tend to show less observed AI research activity — and if so, where are those disparities most consequential?

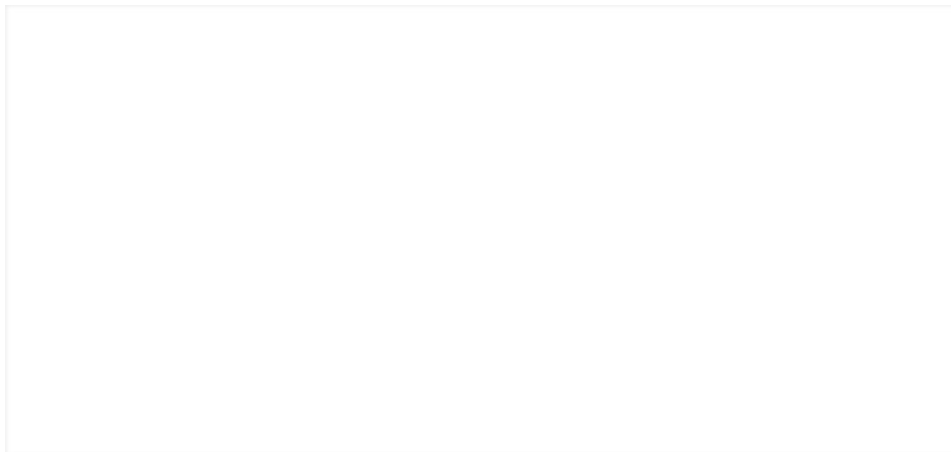


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8,000 cities sized by population and colored by distance to the nearest major cloud compute region. Diamonds mark deployed AWS, Azure, and Google Cloud regions. Data: Natural Earth, hyperscaler public documentation. Built and published using ArcGIS Online.

The map above shows the starting point of the atlas: 8,000 cities, each positioned by its population and colored by its distance to the nearest major AWS, Azure, or Google Cloud region. Grey diamonds mark the cloud regions themselves. The concentration is immediately visible — dense compute corridors run through North America, Western Europe, and East Asia, while large stretches of Africa, Central Asia, and parts of Latin America sit far from any major cloud region.

The sections that follow describe how this atlas was constructed, what the resulting patterns reveal, and why the relationship between compute proximity and observed AI activity is more structured than it first appears.



328 matched AI cities from the OpenAlex scholarly record, sized by recent AI works and colored by distance to the nearest cloud region. The dense cluster across Western Europe and East Asia — and the near-absence of points across Africa and Central Asia — mirrors the access gradient visible in the hero map above. OpenAlex matches are aggregated to unique cities. Built using ArcGIS Online.

When observed AI research activity is overlaid on the accessibility map, the correspondence is striking. Of the 328 cities in the delivered AI overlay, 72 percent sit within 500 km of a major cloud region — compared to just 44 percent of all large cities in the 8,000-city frame. The median AI-linked city is 237 km from its nearest cloud region; the median large city is 657 km away. AI research activity is not distributed in proportion to urban population. It is concentrated in the compute-proximate parts of the global city system.

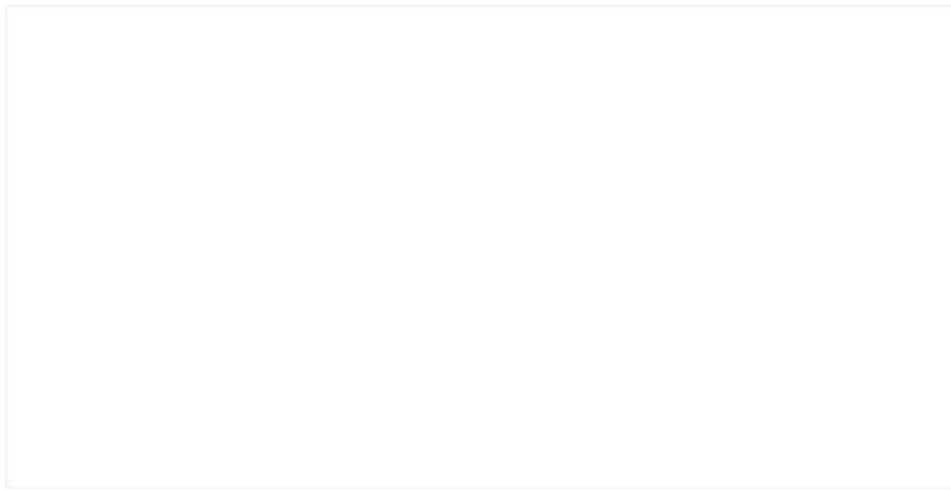
How the atlas works

The atlas is built in three stages using ArcGIS Online and ArcGIS Pro, with supporting data processing in Python.

Stage 1 — A global comparison frame. The analysis begins with the world's 8,000 largest cities by population, drawn from the Natural Earth populated places dataset. This broad frame makes it possible to compare AI-linked cities to the wider global city system rather than focusing only on a small set of well-known technology hubs. Each city is geocoded with coordinates and population estimates.

Stage 2 — Mapping cloud compute infrastructure and measuring access. The atlas maps 60+ major deployed cloud compute regions operated by the three leading hyperscalers — AWS, Azure, and Google Cloud — compiled from each provider's public infrastructure documentation as of early 2026. For each city, the geodesic great-circle distance to its nearest cloud region is calculated, producing a single, transparent, and globally comparable measure of compute accessibility. This distance metric is intentionally simple: it is a proxy for geographic proximity, not a full model of latency, pricing, or GPU availability. Its value lies in comparability across thousands of cities using a consistent methodology.

Stage 3 — Overlaying observed AI research activity and testing the relationship. The final stage overlays observed AI-related research activity derived from the OpenAlex scholarly record, matching recent AI publications to cities by institutional affiliation. This produces a subset of 328 AI-linked cities within the 8,000-city frame. The atlas then tests whether the distance-activity pattern is statistically structured rather than anecdotal, using three methods: Getis-Ord G_i^* hot spot analysis to identify significant spatial clusters of AI activity, Global Moran's I to measure spatial autocorrelation, and two spatial regression specifications — a Gaussian process model and a conditional autoregressive (CAR/GMRF) model — to check whether the negative relationship between distance and AI activity survives controls for city population and geographic structure.



Great-circle distance from each city in the 8,000-city Natural Earth frame to the nearest deployed AWS, Azure, or Google Cloud region. Color encodes distance (dark = close, light = far); marker size scales with city population. The compute-access gradient runs from dense corridors in North America, Europe, and East Asia to structurally distant zones across the Sahel, Central Asia, and interior South America. Sources: Natural Earth, hyperscaler public documentation. Analysis and visualization: ArcGIS Online and ArcGIS Pro.

What the atlas found

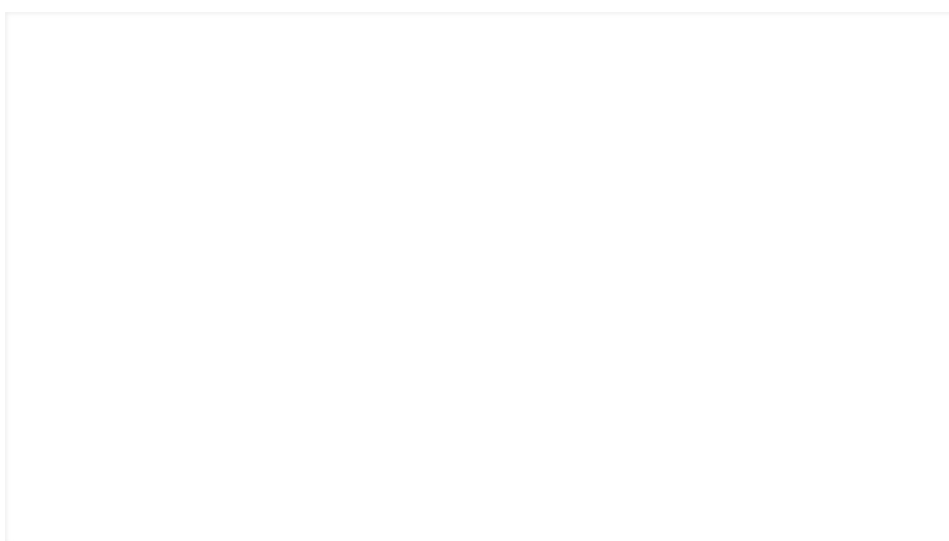
Finding 1: AI-linked cities occupy a fundamentally different part of the compute-access landscape.

The histogram below compares the distance-to-cloud distributions of two populations: all 8,000 large cities in the global frame (blue shading) and the 328 cities matched to recent AI research activity in the OpenAlex record (orange line). Both are plotted on a log-distance scale so that the full global range remains visible while the sharp leftward shift in the AI-linked sample is still clearly resolved.

The separation is immediate — and statistically unambiguous. The AI-linked distribution peaks sharply between 50 and 250 km, while the broader city distribution spreads across a much wider range, extending well past 2,500 km. To confirm that this visual separation reflects a genuine distributional difference rather than an artifact of sample size, three non-parametric tests were applied.

A two-sample Kolmogorov-Smirnov test — which measures the maximum vertical distance between two cumulative distribution functions — returns $D = 0.30$ ($p < 0.001$), confirming that the AI-linked and all-city distance distributions are drawn from different underlying populations. A one-sided Mann-Whitney U test — which tests whether one distribution is systematically shifted below the other — confirms that AI-linked cities are significantly closer to cloud regions than the broader city population ($U = 786,844$; $p < 0.001$). The effect is substantively meaningful, not merely statistically detectable: Cohen's $d = 0.74$, corresponding to a medium-to-large effect size by conventional thresholds.

In concrete terms, 72 percent of AI-linked cities fall within 500 km of a major cloud region, compared to just 44 percent of all large cities. A chi-square test of independence on this 500 km threshold rejects the null hypothesis of equal proportions ($\chi^2 = 103.8$, $df = 1$, $p < 0.001$). Cities producing visible AI research do not simply sit somewhat closer to compute infrastructure on average. They concentrate overwhelmingly inside the densest compute corridors — and are largely absent from the zones farthest from cloud regions.



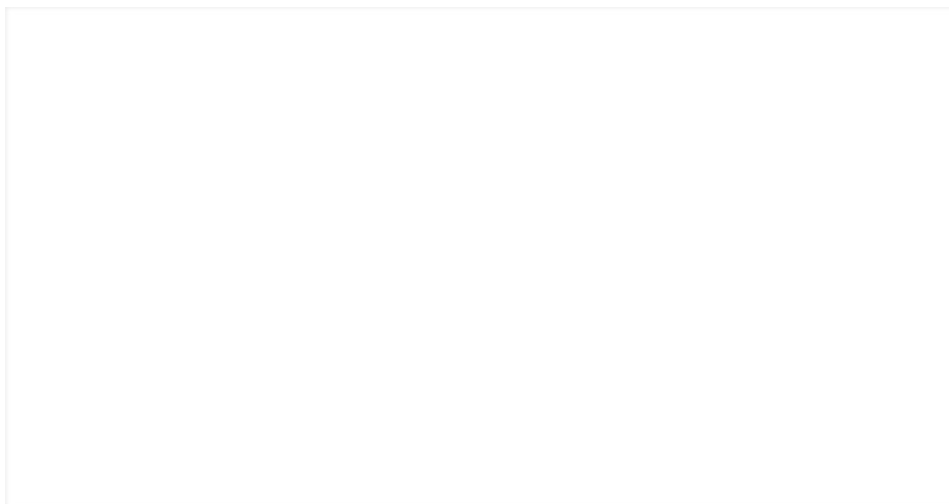
Distance-to-cloud distributions for all large cities ($n = 8,000$, blue shading) and unique matched AI cities ($n = 319$, orange line) on a log-distance axis. Dashed vertical lines mark the respective medians: 248 km for AI-linked cities, 657 km for all large cities. A two-sample Kolmogorov-Smirnov test confirms the distributional shift is significant ($D = 0.29$, $p < 0.001$; Cohen's $d = 0.73$). Equal-width bands on a log scale. Source: Natural Earth city frame, OpenAlex AI-city match.

Finding 2: The concentration sharpens when weighted by research volume.

Finding 1 treats each AI-linked city equally — a city with a single matched publication counts the same as one with hundreds. When the distribution is instead weighted by the number of recent AI works each city produces, the center of gravity shifts further toward cloud infrastructure. The activity-weighted median distance drops to 164 km, compared to **237 km** in the unweighted city-level view.

The intensification is visible in the concentration ratios. Within 500 km of a major cloud region, **72 percent** of AI-linked cities account for **81 percent** of all observed AI works — an excess concentration of approximately **9 percentage points** over what would be expected if research output were distributed uniformly across the AI-linked sample. Within 250 km, **52 percent** of cities produce **62 percent** of all works. The cumulative share line in the figure below illustrates this acceleration: it reaches 81 percent at 500 km and 93 percent at 1,000 km.

It is important to note what this finding does and does not show. The intensification is descriptive: it demonstrates that the cities with the highest research volume are disproportionately located near cloud infrastructure. A Spearman rank correlation between distance and research output within the AI-linked sample alone is weakly negative ($\rho = -0.05$, $p = \mathbf{0.40}$), indicating that once a city is already in the AI-producing set, additional proximity does not strongly predict higher output. The sharper concentration in the weighted view is driven primarily by the fact that the largest AI-producing hubs — cities with hundreds of matched publications — tend to coincide with the world's densest compute corridors, rather than by a smooth gradient within the AI-linked sample.



Activity-weighted distance distribution. Bars show the share of recent AI works (not cities) in each log-distance band; the teal cumulative line tracks total concentration. Dashed line marks the weighted median at 164 km. Within 500 km, 71% of AI cities account for 81% of all observed works. Source: OpenAlex AI-city match, weighted by recent works. Concentration summary only; not a causal estimate.

Finding 3: The pattern is spatially structured, not randomly distributed.

Findings 1 and 2 establish that AI-linked cities sit closer to cloud infrastructure than the broader city system. But a distributional difference alone does not tell us whether the pattern has *spatial structure* — whether nearby cities tend to share similar levels of AI activity, or whether high- and low-activity cities are scattered randomly across the map.

Answering this question requires spatial autocorrelation analysis: a family of methods designed to test whether values observed at one location are statistically related to values at nearby locations.

Two complementary spatial diagnostics are applied here. The first, Global Moran's I, tests for the presence and strength of spatial autocorrelation across the entire matched-city network. The second, Getis-Ord G_i^* , identifies *where* statistically significant clusters of high or low values occur. Together, they answer two distinct questions: is the pattern structured (Moran's I), and where exactly are the clusters (G_i^*)?

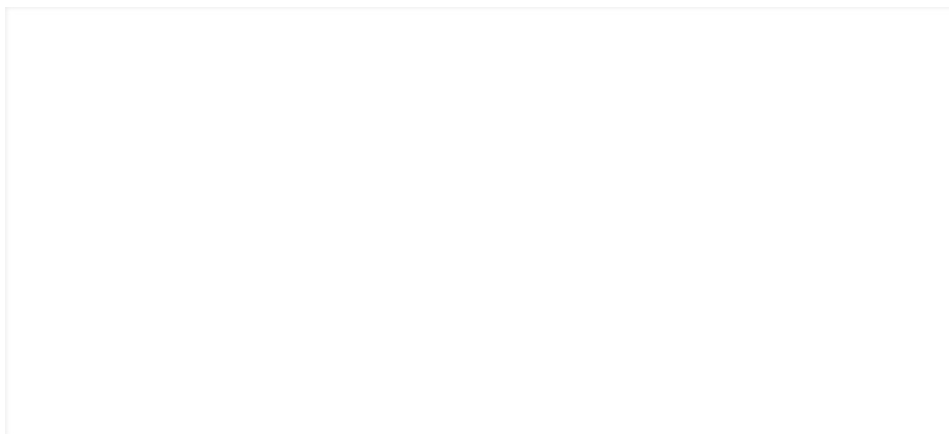
Global Moran's I is a summary statistic that measures the degree to which a variable — in this case, log AI works — is spatially autocorrelated across a set of observations. It compares each city's deviation from the global mean against the deviations of its spatial neighbors, weighted by a spatial weights matrix. A positive Moran's I indicates that similar values tend to cluster together (high near high, low near low); a value near zero indicates spatial randomness; and a negative value indicates dispersion. Applied to the 319 unique matched AI cities, the test returns $I = 0.066$ with a permutation z-score of 2.86 ($p = 0.008$). The autocorrelation is modest in magnitude but statistically significant: cities with high AI activity tend to sit near other high-activity cities, and low-activity cities cluster together as well. The pattern is not random.

*Getis-Ord Gi hot spot analysis** moves from the global summary to local identification. While Moran's I tells us that spatial structure exists, it does not tell us where. Gi^* computes a local statistic for each city by comparing the sum of AI works in its neighborhood against what would be expected under spatial randomness. Cities where the local sum is significantly higher than expected are classified as hot spots; cities where it is significantly lower are classified as cold spots. Significance is assessed at the 95 and 99 percent confidence levels using z-scores derived from the spatial weights structure. Applied to the matched-city network using ArcGIS spatial statistics tools, the analysis identifies 7 statistically significant hot spots and 33 cold spots. The hot spots cluster in East and Southeast Asia — with Macau as the single 99-percent hot spot — while cold spots concentrate in the Americas and peripheral Europe, including Carbondale (United States) as the strongest cold spot.

Why these methods were chosen. Spatial autocorrelation diagnostics are not commonly applied to AI infrastructure or research-output data — most studies in this domain rely on descriptive mapping, regression, or network analysis alone. Moran's I and Gi^* were selected here for three reasons. First,

they are specifically designed to detect whether a spatial pattern is statistically structured or merely visually suggestive — a critical distinction for a project that covers 8,000 cities across diverse geographies. Second, they operate at two complementary scales: Moran's I provides a single whole-map test of spatial dependence, while G_i^* provides city-level cluster assignments that can be mapped and interpreted individually. Third, both methods are implemented in the ArcGIS Spatial Statistics toolbox, making them reproducible within the Esri ecosystem that this project is built on.

Alternative approaches were considered. Kernel density estimation (KDE) could produce a visually similar hot spot surface but does not provide statistical significance tests — it shows where points are dense, not whether that density is greater than expected given the underlying city distribution. DBSCAN or other clustering algorithms could group cities by proximity, but they cluster on location alone and do not incorporate the attribute of interest (AI works). Local indicators of spatial association (LISA), the local decomposition of Moran's I, would provide a valid alternative to G_i^* and would additionally distinguish between high-high clusters, low-low clusters, and spatial outliers. LISA was not used here because G_i^* produces a more interpretable hot/cold classification for a public-interest StoryMap audience, but a LISA analysis would be a natural extension for a more technical report.



Getis-Ord Gi* cluster classification for 319 unique matched AI cities. Color indicates cluster class: hot spot at 99% (1 city), hot spot at 95% (6 cities), not significant (279 cities), cold spot at 95% (23 cities), cold spot at 99% (10 cities). Marker size scales with recent AI works. Global Moran's I = 0.066, $z = 2.86$, $p = 0.008$. Analysis conducted using Getis-Ord Gi* and Global Moran's I in the ArcGIS Spatial Statistics toolbox. The Gi* surface identifies descriptive spatial concentration, not causal treatment effects.

Finding 4: The atlas identifies 1,988 priority cities where compute distance and absent AI activity converge.

Finding 4: The atlas identifies 1,988 priority cities where compute distance and absent AI activity converge.

Findings 1–3 establish that AI-linked cities are disproportionately close to cloud infrastructure and that this pattern is spatially structured. The natural follow-up question is: which cities are on the wrong side of both gaps — far from compute *and* absent from the AI research overlay?

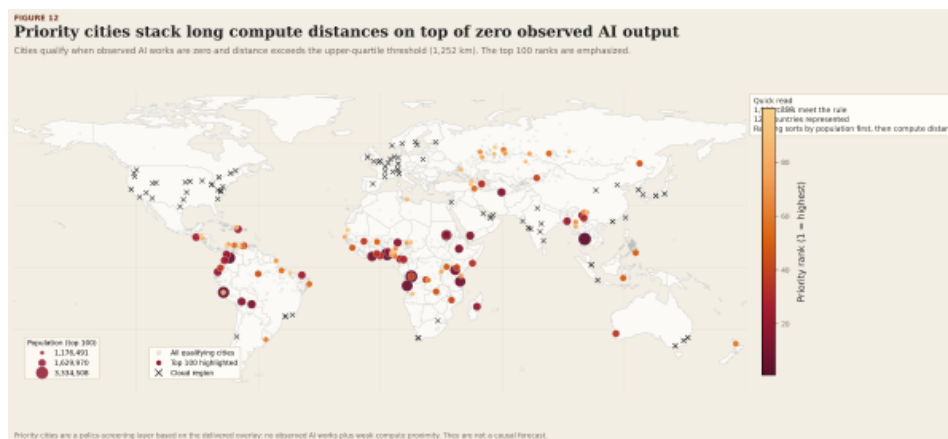
To answer this, the atlas applies a screening rule. A city qualifies as a priority case if it meets two conditions simultaneously: zero observed AI works in the delivered OpenAlex overlay, and a distance to the nearest cloud region that exceeds the upper-quartile threshold of 1,252 km. Cities meeting both criteria are then ranked by population first and compute distance second, so that the largest underserved cities surface at the top of the priority list.

The screen flags 1,988 cities across 125 countries. The concentration is not random — it maps onto the same

geographic structure identified in Findings 1–3. Priority cities cluster heavily in Sub-Saharan Africa, where 468 cities qualify (with a median compute distance of 2,676 km and only 3 cloud regions on the entire continent, all in South Africa), and in Latin America, where 704 cities qualify (median distance 1,298 km, with cloud infrastructure concentrated in Brazil and Chile). Smaller but significant clusters appear in Central Asia, the Middle East, and parts of Southeast Asia.

The top-ranked priority cities are not marginal settlements. They include some of the world's largest and fastest-growing urban centers — cities with millions of residents, active digital economies, and stated ambitions in technology and innovation. Lagos, Kinshasa, Khartoum, Lima, and Bogotá all appear in the top ranks. These are places where the gap between urban scale and compute accessibility may be most consequential for AI participation.

This screening layer transforms the atlas from a descriptive exercise into a practical public-interest tool. For planners, innovation agencies, and international development organizations, the priority list identifies where infrastructure investment in cloud compute could address the starkest mismatches between urban demand and current supply. It is not a causal forecast — it does not predict which cities *will* develop AI activity — but it highlights where structural infrastructure conditions are currently least favorable.



Priority-city screening layer. Cities qualify when observed AI works are zero and distance exceeds the upper-quartile threshold (1,252 km). The top 100 are emphasized. Color encodes priority rank (1 = highest); marker size scales with population among the top 100. 1,988 cities across 125 countries meet the screening criteria. Ranking sorts by population first, then compute distance. This is a policy-screening layer, not a causal forecast.

Analysis and visualization: ArcGIS Online.

Finding 5: The distance-activity relationship survives spatial regression controls.

Findings 1–4 are descriptive: they show that AI-linked cities are closer to cloud infrastructure, that the concentration sharpens with research volume, that the pattern is spatially structured, and that nearly 2,000 cities sit on the wrong side of both gaps. But none of these findings account for the possibility that the distance-activity pattern is driven entirely by city size or broad continental geography. A large city is more likely to host AI research *and* more likely to attract a nearby cloud region — which means population could be a confounding variable. Similarly, cities in Europe and North America may score high on both dimensions simply because of their continental position, not because of a direct distance-activity link.

To test whether the relationship between compute distance and AI activity holds after controlling for these factors, two spatial regression models were estimated: a Gaussian process (GP) model and a conditional autoregressive (CAR/GMRF) model. Both regress log AI works on two predictors: log distance to the nearest cloud region and log city population. The key difference between them lies in how they handle

spatial dependence — the tendency for nearby cities to have correlated outcomes regardless of their individual characteristics.

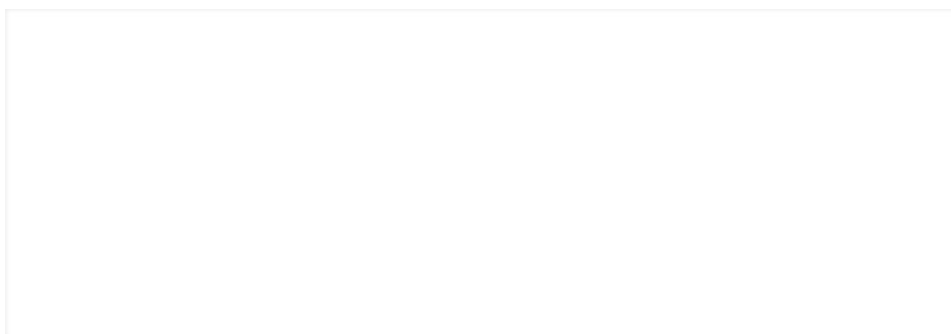
The **Gaussian process model** treats spatial dependence as a smooth, continuous surface. It estimates a latent spatial field across all city locations, allowing the model to absorb broad regional gradients — such as the general tendency for European cities to have both high access and high AI output — before estimating the residual effect of distance and population. The GP specification returns a distance coefficient of **-0.207** (per 1,000 km) and a population coefficient of **+0.279** (log scale). The negative distance sign means that, even after accounting for population and the smooth spatial field, cities farther from cloud infrastructure tend to produce less observed AI research.

The **conditional autoregressive (CAR/GMRF) model** handles spatial dependence differently. Rather than estimating a smooth surface, it models spatial correlation through a neighborhood structure — each city's outcome is partially determined by the outcomes of its spatial neighbors. This is a more aggressive spatial control: it absorbs not only broad regional gradients but also local spillover effects between nearby cities. Under this specification, the distance coefficient attenuates to **-0.052**, while the population coefficient remains positive at **+0.309**. The attenuation is expected and informative: the CAR model absorbs more of the distance signal into the spatial structure, but the sign does not flip. Distance to cloud infrastructure remains negatively associated with AI research output even under the more demanding specification.

The key result across both models is **directional consistency**: the distance coefficient is negative in both, and the population coefficient is positive in both. This is a sign check, not a precise causal effect estimate. The current pipeline stores posterior point estimates but not the full uncertainty intervals needed for formal inferential comparison. What the models confirm is that the descriptive pattern identified in Findings

1–4 is not an artifact of city size or broad geography — it persists as a structured spatial relationship once these factors are controlled for.

This is why the atlas treats compute accessibility as a meaningful spatial correlate of observed AI activity, rather than as a proven causal mechanism. The project does not claim that relocating a city closer to a cloud region would mechanically increase its AI research output. It claims that distance to compute infrastructure is one observable dimension of a broader geographic structure that corresponds with where AI activity is — and is not — currently concentrated.



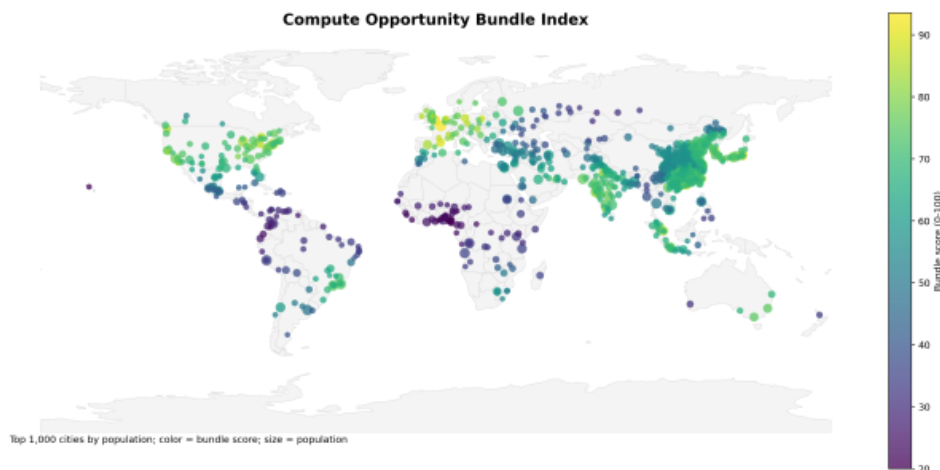
Point estimates from two spatial regression specifications: Gaussian process (GP, teal) and conditional autoregressive (CAR/GMRF, coral). Distance to nearest cloud region (per 1,000 km) is negative in both models (GP: -0.207, CAR: -0.052); log population is positive in both (GP: +0.279, CAR: +0.309). The attenuation from GP to CAR reflects more aggressive spatial dependence absorption. Point estimates only — interval estimates are not available in the current pipeline outputs. These are directional sign checks, not causal effect sizes.

Beyond distance: the infrastructure bundle

Distance to the nearest cloud region is the simplest access measure in the atlas, but it is not the whole story. Cities differ not only in compute proximity, but also in the wider infrastructure bundle that surrounds that proximity.

To make that broader pattern visible, the project adds a composite bundle layer that combines compute access with other enabling conditions such as connectivity, institutional depth, and local infrastructure context.

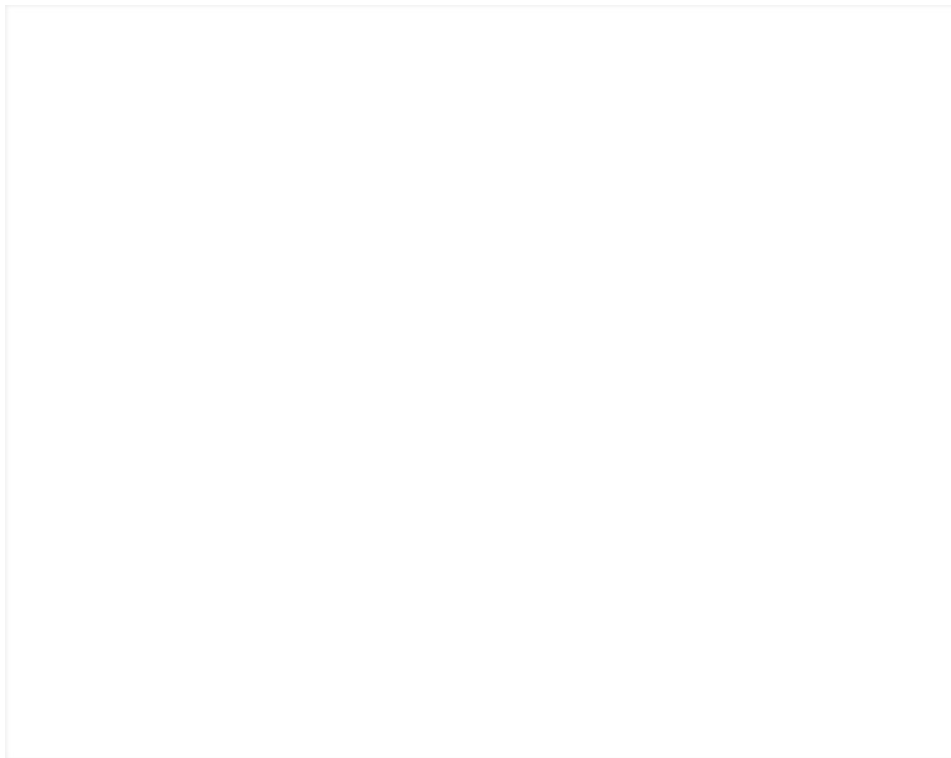
Compute access matters most when it is reinforced by the rest of the infrastructure bundle.



The Compute Opportunity Bundle Index combines compute proximity with wider enabling conditions to show where cities sit inside a stronger or weaker infrastructure bundle.

This map extends the atlas beyond distance alone. Instead of asking only which cities are close to major cloud regions, it asks which cities combine compute access with the broader conditions that help turn access into opportunity.

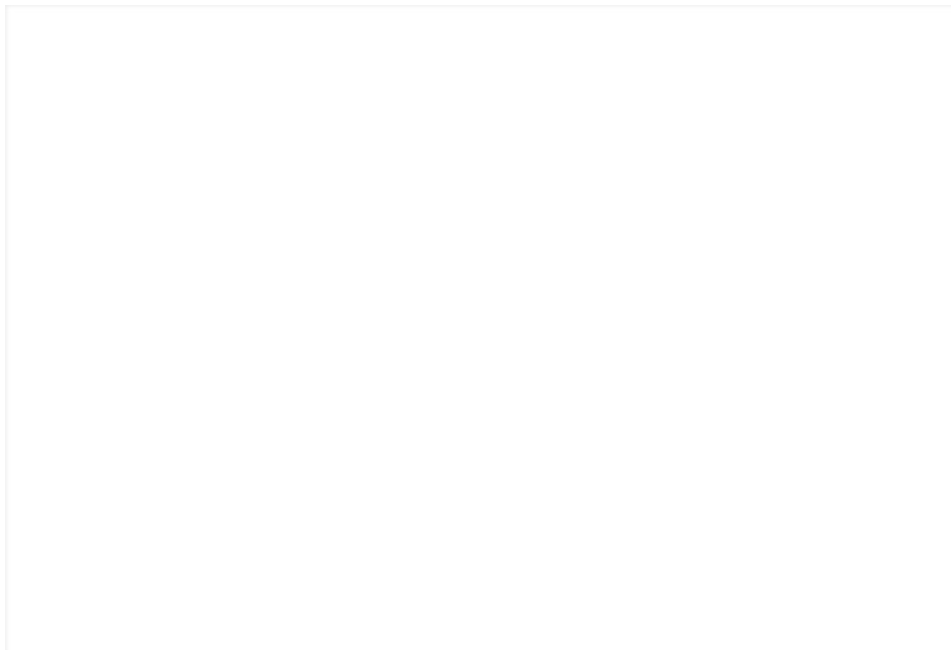
The result is not a replacement for the original atlas. It is a second layer of interpretation that helps distinguish cities with similar distance values but very different surrounding infrastructure.



Distance alone and bundle access do not produce the same geography. Some cities rise or fall once wider infrastructure conditions are included.

This comparison shows why the bundle layer matters. A city can be relatively close to compute but still sit inside a weaker surrounding ecosystem, or farther from compute while remaining stronger on institutions, connectivity, or local infrastructure.

That is why the project does not treat compute distance as destiny. The broader opportunity landscape depends on how compute access combines with the rest of the urban infrastructure system.



The bundle layer surfaces the cities that look strongest overall, as well as the places that depart most clearly from a distance-only ranking.

This final comparison makes the bundle layer concrete. It highlights the cities where strong compute access is reinforced by wider infrastructure advantages, and it also helps identify places where the broader urban system looks stronger or weaker than distance alone would suggest.

In other words, the originality of this project is not just that it maps compute. It is that it treats compute as one visible component of a larger urban opportunity structure.

Why the bundle matters in real places

The bundle layer helps explain why some cities align strongly with the atlas pattern while others depart from it. That is why the next section turns from global patterns to four city cases: one where the full bundle aligns strongly, one where the research overlay understates a broader AI ecosystem, one that outperforms distance alone, and one where infrastructure constraints stack together.

Four cities that explain the pattern

The atlas shows a broad relationship between compute proximity and observed AI activity, but that relationship is not

deterministic. Cities do not rise or lag for one reason alone.

These four cases show why. They reveal that compute access works together with other spatially organized factors such as connectivity, power, institutions, and policy.

Compute is not destiny. It is one layer in the infrastructure bundle that shapes AI opportunity.

Singapore: Near compute / high AI

Singapore is the clearest case in the atlas where the main pattern lines up strongly. It sits extremely close to major cloud infrastructure and also shows high observed AI activity.

This makes Singapore the best example of what the full infrastructure bundle looks like when compute access, connectivity, institutions, and strategy reinforce one another.



Singapore sits close to major cloud infrastructure and inside one of the strongest digital corridors in Asia.

Singapore is not only close to cloud regions. It also sits inside a dense regional connectivity system with strong international links and a role that extends well beyond its small physical size.

In the atlas, this is what alignment looks like: a city that is close to major compute infrastructure and deeply embedded in wider network geography.



At the local scale, cloud infrastructure, institutions, and national strategy reinforce one another.

Singapore's local ecosystem helps explain why it appears as a strong near-compute, high-AI case. Its cloud infrastructure sits alongside major research institutions, an active data-center environment, and a national strategy that treats AI as an economic priority.

The result is not just proximity to compute, but a broader system in which multiple enabling layers reinforce one another.

Takeaway: Singapore shows what happens when compute access, connectivity, institutions, and policy all align.

Seoul: Near compute / low AI

Seoul is one of the most useful exception cases in the project. In the atlas, it appears close to major compute infrastructure, yet relatively weaker in the delivered research overlay than the core high-AI hubs.

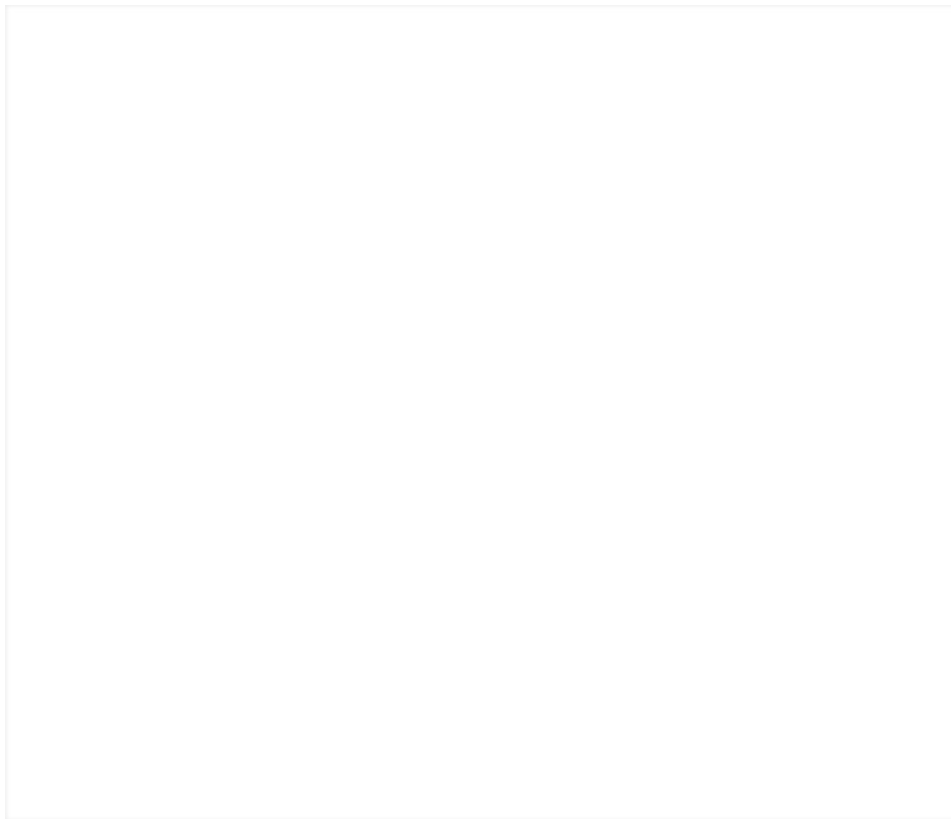
That makes Seoul important because it shows that compute access and measured research output are not the same thing.



Seoul is close to compute and embedded in a dense Northeast Asian infrastructure corridor.

Seoul sits close to major cloud infrastructure and inside a strong regional connectivity corridor. By the access measure alone, it looks like a city that should fall squarely in the high-opportunity group.

Yet in the current research overlay, Seoul is not as prominent as the strongest near-compute, high-AI cases. That tension is exactly why it belongs in the StoryMap.



The broader ecosystem is stronger than the delivered research overlay alone suggests.

Seoul's broader AI ecosystem is stronger than the delivered research overlay alone suggests. This case reminds the reader that publications are only one measure of AI opportunity.

A city can have strong deployment, industry, institutional capacity, and public strategy even when the current research overlay looks comparatively thin.

Ho Chi Minh City: Far compute / high AI

Ho Chi Minh City is the clearest beats-the-pattern case in the atlas. It sits farther from major cloud regions than the high-access hubs, yet still shows substantial observed AI activity.

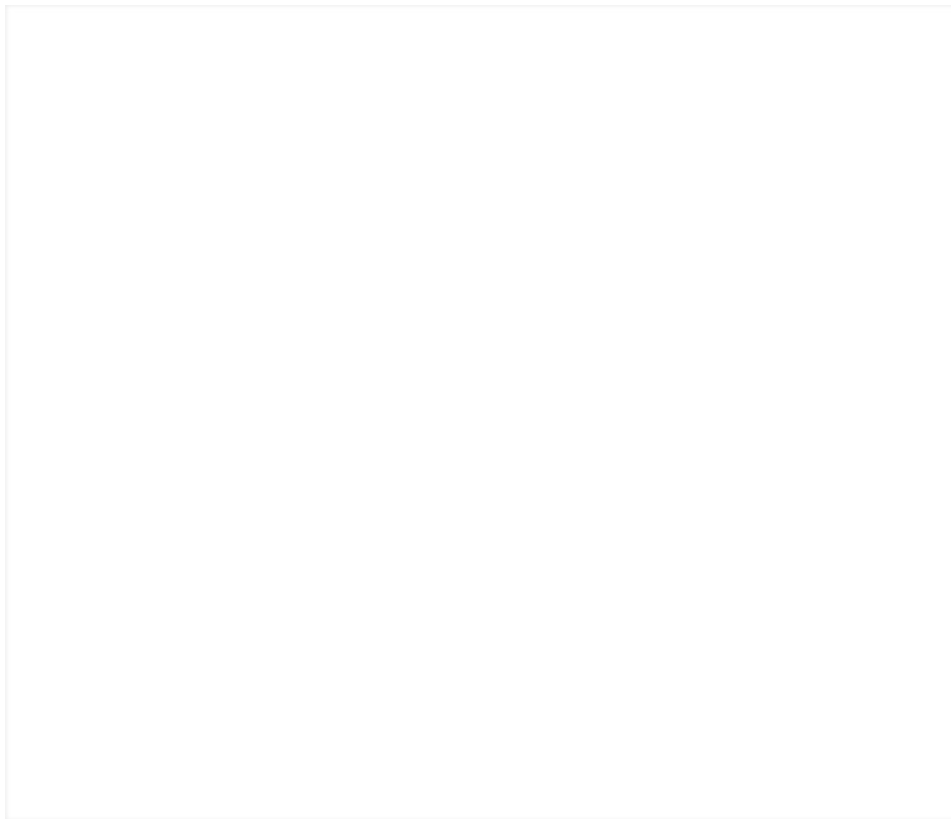
This makes it the best case for showing that distance to compute matters, but does not fully determine what a city can do.



Ho Chi Minh City is farther from major cloud regions, but not disconnected from the wider Southeast Asian digital corridor.

Ho Chi Minh City is farther from hyperscaler infrastructure than the core high-access cities, but it is not geographically isolated. It sits inside a wider Southeast Asian digital corridor and can still connect into stronger regional systems.

This is why the city is so important analytically: it shows that weaker compute proximity does not always translate into weak visible AI activity.



Institutional capacity and broader strategy help explain why the city outperforms distance alone.

Ho Chi Minh City's stronger-than-expected position suggests that compute proximity is only one layer of the story. Institutional capacity, firm activity, and broader national digital strategy can help cities outperform what distance alone would predict.

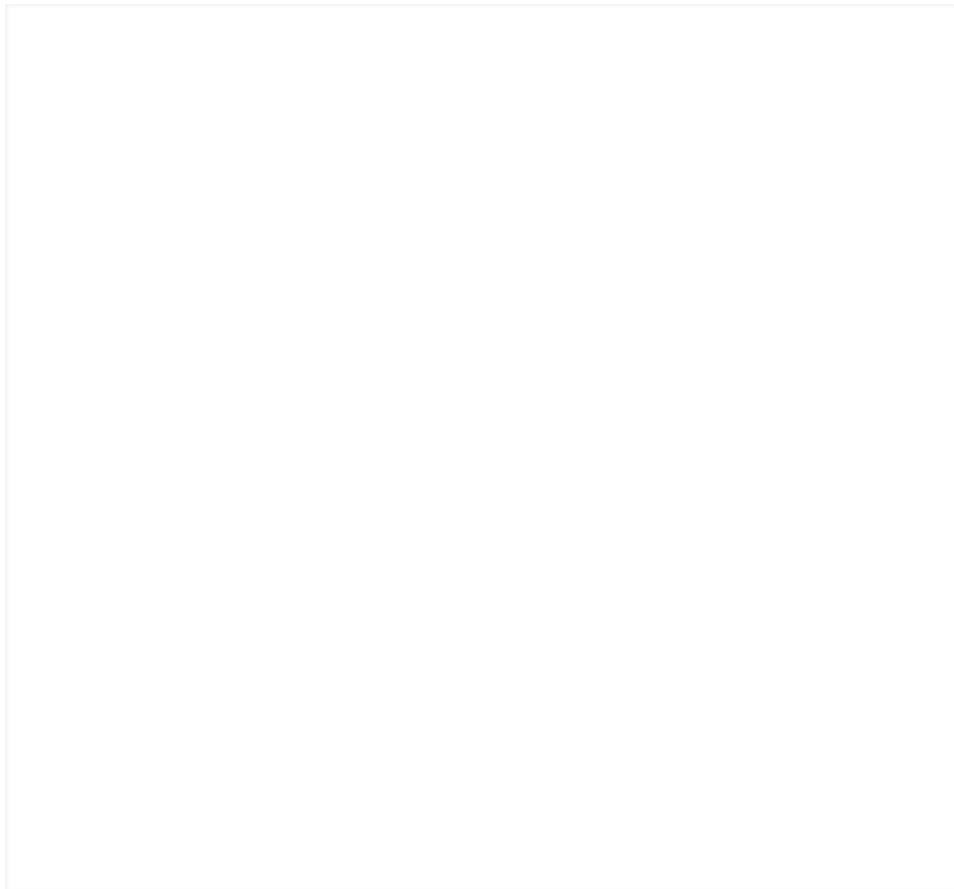
This is the case that most clearly demonstrates why the project needs case studies alongside the atlas.

Takeaway: Ho Chi Minh City shows that strong institutions and regional connectivity can offset weaker compute proximity.

Lagos

Lagos is one of the clearest stacked-disadvantage cases in the atlas. It is a very large urban center, but it remains far from major hyperscaler infrastructure and appears weak in the delivered research overlay.

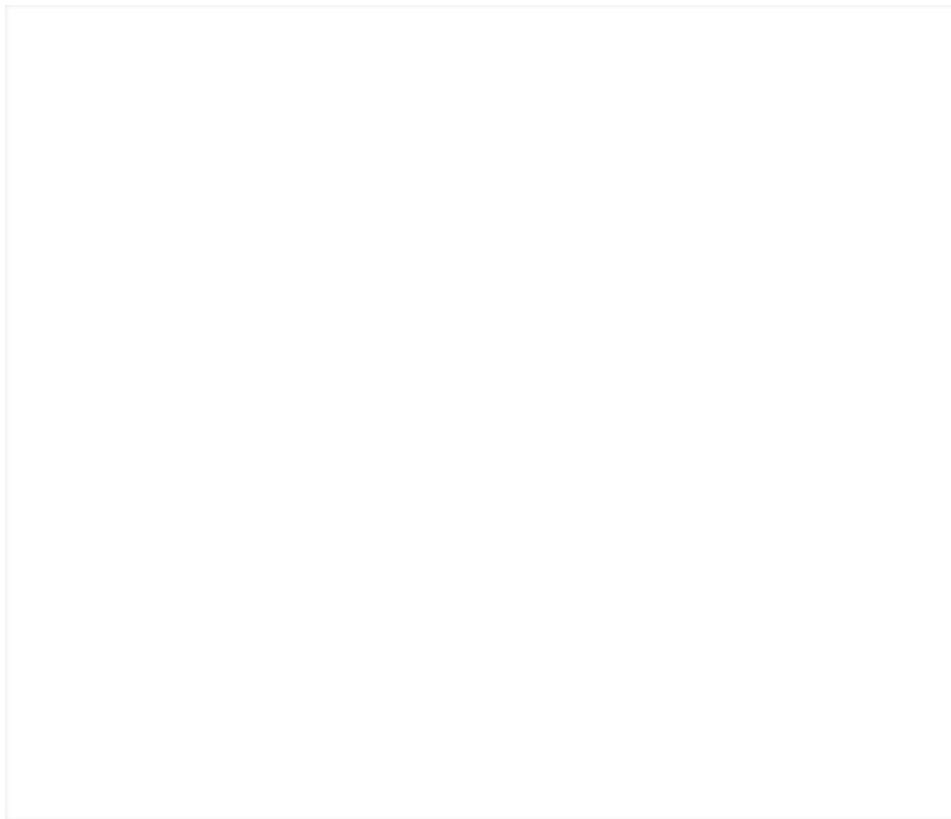
This makes Lagos one of the most important public-interest cases in the project.



Lagos is a major urban market, but it remains relatively far from major cloud infrastructure.

Lagos is not a marginal city. It is one of the largest urban markets in Africa, which makes its distance from major cloud infrastructure analytically important.

In the atlas, Lagos helps show that urban scale alone does not guarantee close access to the compute systems that support modern AI.



Connectivity exists, but infrastructure frictions stack together more heavily here than in the stronger cases.

Lagos is not disconnected from the digital economy. It has visible startup energy, connectivity infrastructure, and local demand.

But this case suggests that compute gaps matter most when they stack together with thinner power reliability, weaker data-center depth, and other infrastructural constraints. That is what makes Lagos such an important priority-city example.

Takeaway: Lagos shows how compute distance can become most consequential when it combines with other infrastructure constraints.

Together, these four cities explain why the atlas is useful and why it must be interpreted carefully.

Singapore shows what infrastructure alignment looks like. Seoul shows that the current research overlay does not capture the whole AI economy. Ho Chi Minh City shows that cities can outperform the distance pattern. Lagos shows how compute gaps can stack with other infrastructure constraints.

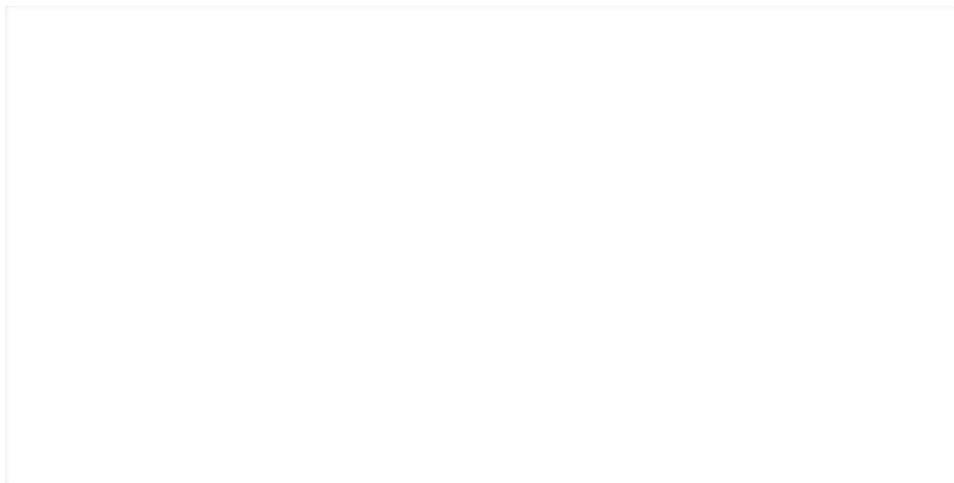
The result is a more mature conclusion: compute is not destiny, but it is part of the infrastructure bundle that shapes AI opportunity.

What this means

The atlas is not a map of who can and cannot do AI. It is a map of one hidden infrastructure layer that helps shape the geography of AI opportunity.

Taken together, the global results and the four city cases suggest a more mature conclusion: compute access matters, but it matters as part of a wider system that also includes connectivity, power, institutions, and policy.

Compute is not destiny. But it is not irrelevant either.

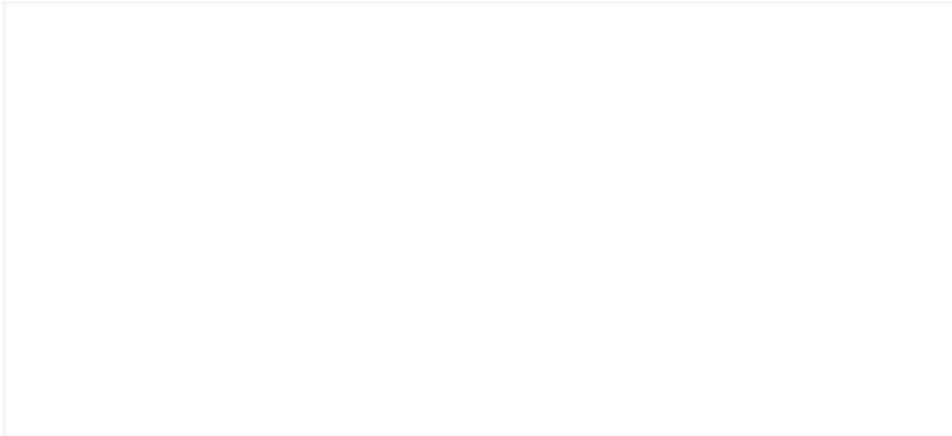


The atlas helps distinguish cities where compute access and AI activity align from cities that beat or lag the broader pattern.

The atlas contributes something simple but important: it makes a hidden infrastructure layer visible.

Public discussion often treats AI as if it depends only on talent, firms, or research. This project shows that proximity to cloud compute is also geographically uneven, and that those uneven access patterns line up in meaningful ways with observed AI activity.

In that sense, the atlas does not replace other explanations. It adds an overlooked one.



The atlas also works as a screening tool by highlighting large cities where compute-access gaps may matter most.

The project is not only descriptive. It is also practical.

By identifying cities where scale, infrastructure, and observed AI activity do not line up neatly, the atlas creates a public-interest tool for thinking about uneven digital opportunity. That makes it useful not only for researchers, but also for planners, universities, innovation agencies, and public-facing audiences trying to understand where AI advantage may be concentrated and where it may still be thin.

The strongest conclusion supported by the evidence is not that cloud proximity mechanically causes AI success everywhere.

The stronger conclusion is that compute is one part of the infrastructure bundle that shapes AI opportunity, and that ignoring its geography leaves part of the story invisible.

The four case studies make this especially clear: some cities align strongly with the atlas pattern, some are underestimated by the current research overlay, some outperform what distance alone would predict, and some face stacked infrastructure constraints.